# Contents

**Complex Analysis for Electrodynamics and Mechanics: a Study Guide** **1**

**CA01: Complex numbers, polar form, roots and the exponential** **5**

**CA02: Analytic functions and the Cauchy–Riemann equations** **8**

**CA03: Elementary functions, the logarithm, $z^a$ and branch cuts** **11**

**CA04: Contour integrals and Cauchy's theorem** **13**

**CA05: Cauchy's integral formula, maximum modulus and harmonic functions** **16**

**CA06: Taylor and Laurent series** **19**

**CA07: Singularities and the residue theorem** **22**

**CA08: Improper integrals, Jordan's lemma and principal values** **25**

**CA09: Capstone, the three Green functions, Wick rotation and the gate** **28**

**Appendix: parked material, the QFT outline (Part 2), vocabulary, conventions and tracking** **31**

# Complex Analysis for Electrodynamics and Mechanics: a Study Guide

*From Ávila's* Variáveis Complexas e Aplicações *to the retarded, advanced and Feynman Green functions*

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**Target.** The complex analysis needed for the electrodynamics and mechanics guides, with one hard deadline: the contour integral that produces the retarded, advanced and Feynman Green functions (EM guide, Module 11). Secondary targets: Kramers–Kronig (EM Module 19) and a clean foundation for the QFT track.

**Starting point, stated honestly.** Complex analysis is your weak spot, and you want to follow this path rather than skim it. So every module has a fast exit (a skip-test), a tier system (Must, Should, Could), a method card for each problem type, ten exercises with expected results, and a blank-page gate.

**Format.** This overview, one file per module, and an appendix.

| File | Content |
|---|---|
| README.md | This overview |
| ca01-complex-numbers.md | Complex numbers, polar form, roots, the exponential |
| ca02-analytic-functions-cauchy-riemann.md | Analytic functions and the Cauchy–Riemann equations |
| ca03-elementary-functions-logarithm.md | Elementary functions, the logarithm, $z^a$, branch cuts |
| ca04-contour-integrals-cauchy-theorem.md | Contour integrals and Cauchy's theorem |
| ca05-cauchy-formula-harmonic-functions.md | Cauchy's integral formula, maximum modulus, harmonic functions |
| ca06-taylor-laurent-series.md | Taylor and Laurent series |
| ca07-singularities-residues.md | Singularities and the residue theorem |
| ca08-improper-integrals-jordan-principal-value.md | Improper integrals, Jordan's lemma, principal values |
| ca09-capstone-three-green-functions.md | Capstone: the three Green functions, Wick rotation, the gate |
| 99-appendix.md | Parked material, the QFT outline (Part 2), vocabulary, conventions, tracking |
| tools/verify_expected_results.py | Script that checks the expected results quoted in the exercises |

Books: **Ávila**, *Variáveis Complexas e Aplicações* (primary); **Churchill and Brown**, *Complex Variables and Applications* (drill problems and English vocabulary); **Stewart** (prerequisites: 16.2, 16.4, Chapter 11).

**Honesty notes.**

1. This replaces my earlier "Part 1" (seven weeks) and the informal "Part 2". Part 2 is now an outline in the appendix and becomes modules after the CA09 gate.
2. Ávila page numbers come from the table of contents you sent. I have not seen Ávila's exercise lists, only the pages where they sit. So each module points you to his exercise sets by page, and adds ten described exercises with expected results. I dropped the edition and publisher I had written before: I had no basis for them.
3. Churchill chapter numbers are from memory for the 8th and 9th editions. Verify them against your copy.
4. The numerical and closed-form expected results are checked by `tools/verify_expected_results.py`. Proof-type exercises say what to show and have no number to check.
5. The time estimates use your real pace (you rebuild each section on paper), not the faster pace behind my first estimate of 25 hours. That is why the totals below are larger.

## 1. The honest accounting

A session is 2 to 3 hours, as in your other guides.

| Path | Sessions | Hours | What you get |
|---|---|---|---|
| Full path | about 33 | about 83 | Everything below, including the harmonic-function and Dirichlet material |
| Deadline path | about 25 | about 63 | Exactly the chain to EM Module 11: CA01 (skip-test), CA02, CA03 (light), CA04, CA05 (Must), CA06, CA07, CA08, CA09 |
| Shortcut lane | 2 | 5 | Two method cards, to get through EM Module 02 if it arrives first (section 3) |

At one session a week the deadline path ends about when EM Module 11 starts (week 23 or so on the full EM path). A few weeks of overlap cost nothing: EM Modules 09 and 10 do not use contours.

## 2. Module list

| # | Module | Ávila pages | Churchill (verify) | Sessions (full / Must) | Priority | Payoff |
|---|---|---|---|---|---|---|
| CA01 | Complex numbers, roots, exponential | 1–33 | 1 | 2 / 1 | skip-test | $e^{i\theta}$, roots of unity, point sets |
| CA02 | Analytic functions, Cauchy–Riemann | 34–60 | 2 | 4 / 3 | core | Analyticity as a strong condition; $\lvert f'\rvert^2$ is a Jacobian |
| CA03 | Elementary functions, logarithm, branch cuts | 61–74 | 3 | 2 / 1 | should | Multivaluedness; why cuts exist |
| CA04 | Contour integrals, Cauchy's theorem | 75–101 | 4 | 4 / 4 | core, star | Green's theorem becomes Cauchy; the master integral |
| CA05 | Cauchy's formula, maximum modulus, harmonic functions | 101–117 | 4 | 5 / 2 | core (Must part) | Values from boundaries; Earnshaw |
| CA06 | Taylor and Laurent series | 118–150 | 5 | 6 / 4 | core, star | Laurent is how residues are defined |
| CA07 | Singularities and residues | 151–161 | 6 | 4 / 4 | core, two stars | Residue theorem |
| CA08 | Improper integrals, Jordan's lemma, principal values | 161–169, 173–174 | 7 | 4 / 4 | core, three stars | Closing contours; indentation; Kramers–Kronig |

| # | Module | Ávila pages | Churchill (verify) | Sessions (full / Must) | Priority | Payoff |
|---|---|---|---|---|---|---|
| CA09 | Capstone and gate | none | 7 | 2 / 2 | core, three stars | The three Green functions |

The three-star modules carry the guide. If time collapses, protect CA07 to CA09 at the cost of anything else.

**Dependency map.**

```
CA01 -> CA02 -> CA03 (light)
          \
           -> CA04 -> CA05 -> CA06 -> CA07 -> CA08 -> CA09
```

CA06 needs Stewart Chapter 11 (series of constants, radius of convergence). Its skip-test tells you whether the one-session patch is needed.

## 3. How this plugs into the other guides

| Here | Needed by | Note |
|---|---|---|
| CA04 (contours, ML-estimate) | EM Module 02 (recipe), EM Module 11 | Needs Stewart 16.2 and 16.4, which you have |
| CA07, CA08 (residues, closing contours) | EM Module 02 exercises 6 to 8 | See the shortcut lane below |
| CA08, CA09 | EM Module 11 (hard deadline), Module 19 (Kramers–Kronig) | |
| CA06 (Laurent, binomial series) | EM Modules 06 and 07 (Legendre generating function) | |
| CA05 (harmonic functions, maximum modulus) | EM Modules 04 and 05 | Read after Jackson's uniqueness proof, as a second proof |
| CA09 | Zettili (scattering, propagators) | |
| Part 2 (appendix) | Mechanics guide Module 16 (stationary phase), QFT | Not before the CA09 gate |

**Shortcut lane.** If EM Module 02 arrives before you reach CA07 and CA08, do only this: read the residue method card in CA07 and the "closing a contour" card in CA08, do EM Module 02's exercises 6 and 7, then go back to the sequence here. Do not skip CA08 itself.

## 4. The routine for a contour problem (the five steps)

1. **Draw.** Plot the poles and the contour in the plane. Mark which half-plane the contour closes in.
2. **State what happens at infinity.** For a rational function use the ML-estimate. For $e^{iax}g(x)$ use Jordan's lemma, and read the sign of $a$ to choose the half-plane.
3. **Locate the singularities**, including where $\pm i\epsilon$ puts each pole. Decide whether each lies inside, outside or on the contour.
4. **Compute residues** with the method card. Keep symbols until the end.
5. **Check.** Test a symmetric case, a limit ($x \to 0$, $\epsilon \to 0$), and one number you know. Run the numerical integral if the answer is a closed form.

**Stuck protocol.** After 20 minutes without progress, write down which step you are stuck on (never "I don't know how to do the problem"). Look only at the next line of the solution or the method card, then close the book and continue. Log the step in `notes/error_log.md` and redo the problem two days later. If the same step appears three times, redo the relevant module drill.

## 5. Schedule and tiers

Same rules as your other guides. A week off costs nothing. After a long gap, use the re-entry protocol: redo the last two derivations of the previous module from a blank page. A 12-hour week pulls the next module's first session forward. Inside each module, tiers: **Must** (core derivations, exercises 1 to 5), **Should** (6 to 8), **Could** (9 to 10 and the extension). Dropping to Must always leaves a coherent module.

Never skip: CA04's Green's-theorem proof, CA07 and CA08 in full, and the CA09 gate.

## 6. Ground rules

1. Derive before you read the answer.
2. One cheat sheet per module, one page.
3. Keep `notes/error_log.md` and a one-line weekly log (hours, sections, exercises).
4. Do Ávila's own exercise sets first (pages are listed in each module; his answers are in the book). The ten exercises in each module are the extra with expected results.
5. Numerical checks are for checking, not a project. `tools/verify_expected_results.py` shows how.

## 7. What this guide leaves out, and where it goes

Branch cuts inside integrals (keyhole contours), Riemann surfaces, analytic continuation, the gamma function, saddle-point methods, fluid dynamics, and conformal mapping. The appendix lists each with its Ávila pages and the cost of restoring it.

## 8. How this connects to your other work

- **EM guide:** table in section 3.
- **Mechanics guide:** no dependency before Module 16, where stationary phase appears (Part 2).
- **Linear algebra and Fourier path (informal):** Figueiredo Chapter 6 (Fourier transform, convolution) is the real-variable side of CA08 and CA09.
- **Conventions** (Fourier transform, time dependence $e^{-i\omega t}$): appendix, section 4. They match EM Module 02.

# CA01: Complex numbers, polar form, roots and the exponential

**Books:** Ávila Chapter 1, pp. 1 to 33 (exercise sets on pp. 7, 11, 15, 19, 24, 31; answers on pp. 12, 20, 25, 33). Churchill Chapter 1 for extra drills and the English vocabulary. **Time:** 2 sessions (1 if you pass the skip-test). **Tier note:** this is recall, not learning. Take the skip-test first.

## Why this matters

Everything in the guide is built on three facts from this chapter: a complex number is a point in the plane, multiplication is a rotation plus a scaling, and the exponential turns addition into multiplication. Polar form $z = re^{i\theta}$ is the working form for the whole guide and for plane waves in Jackson. The $n$ roots of a number are your first meeting with multivaluedness, and the periodicity $e^{z+2\pi i} = e^z$ is the reason the logarithm will need branch cuts in CA03. The point-set vocabulary (open, connected, simply connected) is the hypothesis of Cauchy's theorem in CA04.

## Skip-test (40 minutes, no notes)

1. Write $\dfrac{1+i}{2-i}$ in the form $a+bi$.
2. Write $-1+i\sqrt{3}$ in polar form.
3. Find all fourth roots of $-16$.
4. Show $e^{z+2\pi i} = e^z$ and find all $z$ with $e^z = -1$.
5. Describe the set $\{z : \ |z-1| < |z+1|\}$.
6. Give one example each of a set that is open, closed, connected, and not simply connected.

Expected: (1) $\frac{1+3i}{5}$; (2) $2e^{2\pi i/3}$; (3) $\sqrt{2}(\pm 1 \pm i)$; (4) $z = i\pi(2k+1)$; (5) the right half-plane $\operatorname{Re} z > 0$; (6) an annulus $1 < |z| < 2$ is open, connected, not simply connected.

## Prerequisites

Stewart on the plane and trigonometry. Nothing else.

## Detailed topics (checklist)

**[Must]** $\mathbb{C}$ as $\mathbb{R}^2$ with a product; $\mathbb{R}$ as a subfield; modulus, conjugate, $z\bar{z} = |z|^2$, $1/z = \bar{z}/|z|^2$; the triangle inequality and its reverse form.

**[Must]** Polar form $z = re^{i\theta}$; the argument is multivalued, $\arg z = \theta + 2\pi k$; product and quotient formulas; De Moivre's formula.

**[Must]** $n$-th roots: exactly $n$, equally spaced on a circle; the roots of unity; primitive roots; their sum is zero for $n \geq 2$.

**[Must]** The exponential: $e^z = e^x(\cos y + i \sin y)$, $e^{z+w} = e^z e^w$, never zero, periodic with period $2\pi i$.

**[Should]** Point sets: neighbourhood, open, closed, boundary, bounded, connected (by polygonal paths), domain, simply connected (informally, no holes; formally, every loop shrinks to a point, as in your Stewart homotopy discussion). Name the Jordan curve theorem.

**[Could]** The point at infinity and the Riemann sphere (it returns in CA06, "regularity at infinity").

## Method cards

**Converting forms.** Cartesian to polar: $r = |z|$, $\theta = \operatorname{atan2}(y, x)$. Check the quadrant, not just $\tan\theta$.

**$n$-th roots of $w = \rho e^{i\varphi}$.** (1) Write $w$ in polar form. (2) The roots are $\rho^{1/n} e^{i(\varphi+2\pi k)/n}$ for $k = 0, \ldots, n-1$. (3) Plot them: $n$ equally spaced points. (4) Check by raising one to the $n$-th power.

**Describing a set.** Write $z = x + iy$, square to remove moduli, complete the square. A condition on $|z-a|$ is a disk or a half-plane; a condition on $\operatorname{Re}(1/z)$ is a disk after completing the square.

## Reading plan

| Session | Task |
|---|---|
| A1 | Skip-test. If it fails, read Ávila pp. 1 to 20 (plane, modulus, polar form, De Moivre, roots). Derive item 1 below |
| B1 | Exercises 1 to 5 and the Ávila sets on pp. 7, 11, 15, 19 |
| A2 | Ávila pp. 21 to 33 (exponential, point sets). Derive items 2 and 3 |
| B2 | Exercises 6 to 10, the Ávila sets on pp. 24 and 31, and the cheat sheet |

## Derive from scratch

1. De Moivre's formula, and the product formula for the polar form.
2. The roots of unity: that they are $n$ points, and that their sum is zero.
3. $e^{z+w} = e^z e^w$ from the definition $e^z = e^x(\cos y + i \sin y)$.

## Exercises (with expected results)

1. **(Must)** Compute $(1+i)^{10}$. *Expected:* $32i$.
2. **(Must)** Show the product of all $n$-th roots of unity is $(-1)^{n+1}$. *Expected:* this holds, e.g. $-1$ for $n=2$ and $+1$ for $n=3$.
3. **(Must)** Solve $z^3 = 8i$. *Expected:* $\sqrt{3}+i,\ -\sqrt{3}+i,\ -2i$.
4. **(Must)** Prove the parallelogram law $|z+w|^2 + |z-w|^2 = 2(|z|^2 + |w|^2)$. *Expected:* expand with $|u|^2 = u\bar{u}$.
5. **(Must)** Use De Moivre to express $\cos 3\theta$ and $\sin 3\theta$ in powers of $\cos\theta$ and $\sin\theta$. *Expected:* $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta$; $\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta$.
6. **(Should)** Sum $\sum_{k=0}^{n} \cos k\theta$ as the real part of a geometric series. *Expected:* $\frac{1}{2} + \dfrac{\sin\left((n+\frac{1}{2})\theta\right)}{2\sin(\theta/2)}$. This is the Dirichlet kernel, and it returns in the Fourier series of your linear algebra path.
7. **(Should)** Find all $z$ with $e^z = 1+i$. *Expected:* $z = \frac{1}{2}\ln 2 + i(\frac{\pi}{4} + 2\pi k)$.
8. **(Should)** Describe $\{z \neq 0 : \operatorname{Re}(1/z) > \frac{1}{2}\}$. *Expected:* the open disk $|z-1| < 1$ (the point 0 lies on its boundary and is excluded).
9. **(Could)** For a primitive $n$-th root $\omega$, show $\prod_{k=1}^{n-1}(1-\omega^k) = n$. *Expected:* $n$ (for $n=5$ and $n=7$ this is checked numerically). Hint: factor $z^n - 1$ and differentiate at $z=1$.
10. **(Could)** Classify as open, closed, connected, simply connected: (a) $1 < |z| < 2$; (b) $\{|z| \leq 1\} \cup \{|z-3| \leq 1\}$; (c) $\operatorname{Im} z > 0$; (d) $\mathbb{C} \setminus [0, \infty)$. *Expected:* (a) open, connected, not simply connected; (b) closed, not connected; (c) open, connected, simply connected; (d) open, connected, simply connected (the slit plane, which will be the domain of the principal logarithm).

## Optional extension

Show that $z \mapsto 1/z$ sends circles and lines to circles and lines (use exercise 8 as a model). It is the first conformal map you will meet and it returns in electrostatics by inversion.

## Pitfalls

- Forgetting that arg is defined only up to $2\pi k$.
- Taking the argument of a number in the second or third quadrant from $\arctan(y/x)$ without fixing the quadrant.
- Believing a complex number has one $n$-th root.
- Treating $e^z$ as one-to-one. It is not, and that is the whole story of CA03.

## Gate questions

1. Why is multiplication in the plane a rotation plus a scaling?
2. Why does a nonzero complex number have exactly $n$ distinct $n$-th roots?
3. What does the periodicity of $e^z$ force the logarithm to be?

## One-page cheat sheet contents

Polar form and De Moivre; the root recipe; $e^z$ in Cartesian form; the point-set definitions; the simply connected examples.

## Where this is used

CA03 (logarithm), CA04 (hypotheses of Cauchy's theorem), Jackson's plane waves $e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$.

# CA02: Analytic functions and the Cauchy–Riemann equations

**Books:** Ávila Chapter 2, pp. 34 to 60 (exercise sets on pp. 36, 42, 47, 52, 62; answers and hints on pp. 43, 48, 53, 74). Churchill Chapter 2. **Time:** 4 sessions (Must: 3). **Tier note:** if you can derive the Cauchy–Riemann equations and the conjugate of a harmonic function cold, pass the skip-test and do only exercises 7 and 10.

## Why this matters

Complex differentiability asks for the same limit from every direction in the plane, not just from the left and right. That single demand is much stronger than real differentiability. It forces the Cauchy–Riemann equations, which force $u$ and $v$ to satisfy Laplace's equation. So every analytic function hands you two solutions of $\nabla^2\Phi = 0$, and their level curves are orthogonal. That is why complex analysis solves two-dimensional electrostatics, and it is the same orthogonality you checked numerically in your Exercise C1. The geometric reading, a rotation and a scaling by $|f'|$, shows that the Jacobian of an analytic map is $|f'|^2$. That is the Jacobian from your Stewart 15.9 work, now with a name for its value.

## Skip-test (40 minutes, no notes)

1. Define the complex derivative and explain why the limit must not depend on the direction of $h$.
2. Derive the Cauchy–Riemann equations from that requirement.
3. Show $f(z) = \bar{z}$ is nowhere differentiable.
4. Show $|z|^2$ is differentiable only at 0, and analytic nowhere.
5. Find the harmonic conjugate of $u = x^3 - 3xy^2$.
6. State the polar form of the Cauchy–Riemann equations.

## Prerequisites

CA01. Stewart 14.3 (partial derivatives) and 14.5 (the chain rule).

## Detailed topics (checklist)

**[Must]** A function $f : \mathbb{C} \to \mathbb{C}$ as a pair $u(x, y)$, $v(x, y)$; limits and continuity (the same $\epsilon$-$\delta$ as in $\mathbb{R}^2$); the rules for limits.

**[Must]** The derivative $f'(z_0) = \lim_{h\to 0} \frac{f(z_0+h)-f(z_0)}{h}$; the rules of differentiation are the real ones.

**[Must]** Cauchy–Riemann: $u_x = v_y$ and $u_y = -v_x$. Necessary for differentiability; sufficient if the partials are continuous. Then $f' = u_x + iv_x$. Polar form: $u_r = \frac{1}{r}v_\theta$ and $v_r = -\frac{1}{r}u_\theta$.

**[Must]** Analytic (holomorphic) on an open set. "Analytic at a point" means analytic in a neighbourhood of it, which is why a function can be differentiable at one point and analytic nowhere. An entire function is analytic on all of $\mathbb{C}$.

**[Should]** Geometric reading: $f'(z_0) \neq 0$ rotates by $\arg f'$ and scales by $|f'|$, so the map preserves angles (conformal). The Jacobian determinant of $(x, y) \mapsto (u, v)$ is $u_x v_y - u_y v_x = |f'|^2$.

**[Should]** Harmonic functions: $\nabla^2 u = \nabla^2 v = 0$. For now assume $u, v$ have continuous second partials. CA05 proves that analytic functions are infinitely differentiable, which removes the assumption.

**[Could]** The Wirtinger form: $f$ is analytic exactly when $\partial f/\partial \bar{z} = 0$, i.e. $f$ does not depend on $\bar{z}$.

## Method cards

**The Cauchy–Riemann test.** (1) Write $f = u + iv$ with $u, v$ in $x, y$. (2) Compute $u_x, u_y, v_x, v_y$. (3) Check $u_x = v_y$ and $u_y = -v_x$. (4) Check the partials are continuous (then CR is sufficient). (5) State where CR holds; the set must be open for "analytic".

**Harmonic conjugate.** (1) Check $\nabla^2 u = 0$. (2) Integrate $v_y = u_x$ with respect to $y$, adding an unknown function $g(x)$. (3) Fix $g$ from $v_x = -u_y$. (4) Check the result is analytic: it should be a polynomial or exponential in $z$.

## Reading plan

| Session | Task |
|---|---|
| A1 | Ávila pp. 34 to 49 (functions, limits, continuity). Skim the limit properties |
| B1 | Exercises 1 to 3 and the Ávila sets on pp. 36 and 42 |
| A2 | Ávila pp. 49 to 57 (analytic function, differentiation rules, Cauchy–Riemann). Derive item 1 below |
| B2 | Exercises 4 to 6 and the Ávila sets on pp. 47 and 52 |
| A3 | Ávila pp. 57 to 60 (polar CR, geometric interpretation). Derive items 2 and 3 |
| B3 | Exercises 7 to 10, the Ávila set on p. 62, and the cheat sheet |

## Derive from scratch

1. The Cauchy–Riemann equations from the direction-independence of the limit (approach along the real axis, then the imaginary axis).
2. $\nabla^2 u = 0$ from CR.
3. $\det \partial(u, v)/\partial(x, y) = |f'|^2$.

## Exercises (with expected results)

1. **(Must)** Show $f(z) = \bar{z}$ is nowhere complex-differentiable. *Expected:* $u = x$, $v = -y$ gives $u_x = 1 \neq v_y = -1$.
2. **(Must)** Show $f(z) = |z|^2$ is differentiable only at $z = 0$, with $f'(0) = 0$, and analytic nowhere. *Expected:* CR requires $2x = 0$ and $2y = 0$.
3. **(Must)** Check that $z^2$ satisfies CR and that $u_x$ equals $\operatorname{Re} f'$. *Expected:* $u = x^2 - y^2$, $v = 2xy$, $f' = 2z$.
4. **(Must)** Find a harmonic conjugate of $u = x^3 - 3xy^2$. *Expected:* $v = 3x^2y - y^3 + C$ (so $f = z^3$).
5. **(Must)** Show $u = e^x \cos y$ is harmonic and find its conjugate. *Expected:* $v = e^x \sin y$ (so $f = e^z$).
6. **(Should)** Verify the polar CR equations for $z^2 = r^2 e^{2i\theta}$. *Expected:* $u_r = 2r \cos 2\theta = v_\theta / r$.
7. **(Should)** Show the Jacobian of an analytic map is $|f'|^2$, and check it for $f = z^3$ at $z = 1 + i$. *Expected:* 36 (since $f' = 3z^2 = 6i$ there).
8. **(Should)** Show that an analytic $f$ with constant $|f|$ is constant. *Expected:* differentiate $u^2 + v^2 = c$ and combine with CR.
9. **(Could)** Where is $f(z) = x^2 + iy^2$ differentiable? *Expected:* only on the line $y = x$, where $f' = 2x$; analytic nowhere.
10. **(Could)** Show that if $f$ and $\bar{f}$ are both analytic on a domain, $f$ is constant. *Expected:* the two sets of CR equations force all four partials to vanish.

## Optional extension

Show that the level curves of $u$ and $v$ are orthogonal wherever $f' \neq 0$ (use $\nabla u \cdot \nabla v = 0$). For $f = z^2$ this is $\nabla u = (2x, -2y)$ and $\nabla v = (2y, 2x)$.

## Pitfalls

- Checking CR at one point and calling the function analytic.
- Forgetting that "analytic" needs a neighbourhood, so isolated points of differentiability do not count.
- Mixing $u_y = -v_x$ with $u_y = v_x$.
- Using polar CR without the factor $1/r$.

## Gate questions

1. Why is complex differentiability much stronger than real differentiability?
2. Why does CR force $u$ and $v$ to be harmonic?
3. Why is the Jacobian of an analytic map $|f'|^2$ and not an arbitrary determinant?

## One-page cheat sheet contents

CR (Cartesian and polar); $f' = u_x + iv_x$; the conjugate recipe; the Jacobian identity; the list of non-analytic examples ($\bar{z}$, $|z|^2$, $\operatorname{Re} z$).

## Where this is used

CA04 (Cauchy's theorem uses CR), CA05 (harmonic functions), EM Modules 04 and 05 (uniqueness and Laplace's equation), conformal electrostatics (appendix).

# CA03: Elementary functions, the logarithm, $z^a$ and branch cuts

**Books:** Ávila Chapter 2, pp. 61 to 74 (exercise sets on pp. 62, 64, 73; answers and hints on p. 74). Churchill Chapter 3. **Time:** 2 sessions (Must: 1). **Tier note:** this is a "Should" module. In the deadline path read it once, do exercises 1 to 3, and come back before the Part 2 material.

## Why this matters

Because $e^z$ is periodic, it has no inverse, so $\log z$ cannot be single-valued. To get a function you cut the plane and pick a branch. The same issue makes $z^a$, $\sqrt{z^2-1}$ and every function with a fractional power multivalued. Branch cuts reappear in contour integration (keyhole contours, Part 2) and in QFT, where a cut in a propagator marks the threshold for multi-particle production. In this guide you need only the vocabulary and the basic recipe, so that a cut in an integrand is never a surprise.

## Skip-test (40 minutes, no notes)

1. Find all values of $\log(-1)$ and the principal value $\mathrm{Log}(-1)$.
2. Compute $i^i$ (principal value).
3. Solve $\sin z = 2$.
4. State the domain on which the principal $\mathrm{Log}\, z$ is analytic and give its derivative.
5. Explain, in two sentences, why $\sqrt{z}$ has a branch point at 0.

## Prerequisites

CA01, CA02.

## Detailed topics (checklist)

**[Must]** The exponential and its periodicity; $\sin z, \cos z$ via exponentials; they are unbounded (unlike the real ones); the hyperbolic functions.

**[Must]** The logarithm: $\log z = \ln|z| + i(\arg z + 2\pi k)$; the principal value $\mathrm{Log}\, z$ with $-\pi < \mathrm{Arg}\, z \leq \pi$; analytic on $\mathbb{C} \setminus (-\infty, 0]$ with $(\mathrm{Log}\, z)' = 1/z$.

**[Must]** Powers: $z^a = e^{a \log z}$, multivalued in general; $z^{1/2}$ has two values.

**[Should]** Branch point versus branch cut: the point is intrinsic (a loop around it changes the function), the cut is a choice. $\mathrm{Log}(z_1 z_2) \neq \mathrm{Log} z_1 + \mathrm{Log} z_2$ in general.

**[Should]** Inverse trigonometric functions as logarithms.

**[Could]** The logarithm as a transformation and its inverse (Ávila p. 67).

## Method cards

**Solving $e^z = w$.** $z = \ln|w| + i(\arg w + 2\pi k)$.

**Solving $\sin z = w$.** Put $\zeta = e^{iz}$: $\zeta - 1/\zeta = 2iw$ is a quadratic in $\zeta$. Solve, then take $z = -i \log \zeta$.

**Choosing a branch.** (1) Find the branch points (where the argument of the expression winds). (2) Choose a cut joining them, or running to infinity. (3) Require continuity on any loop that does not cross the cut. (4) Fix the value at one point.

## Reading plan

| Session | Task |
|---|---|
| A1 | Ávila pp. 61 to 70 (exponential, trigonometric and hyperbolic functions, the logarithm, $z^a$). Derive items 1 and 2 |
| B1 | Exercises 1 to 5 and the Ávila sets on pp. 62 and 64 |
| A2 | Ávila pp. 70 to 73 (inverse trigonometric functions). Derive item 3 |
| B2 | Exercises 6 to 10, the Ávila set on p. 73, and the cheat sheet |

## Derive from scratch

1. $\log z = \ln|z| + i(\arg z + 2\pi k)$ from $e^w = z$.
2. $(\operatorname{Log} z)' = 1/z$ on the cut plane (use the inverse-function rule).
3. $\sin z = w$ solved through the quadratic in $e^{iz}$.

## Exercises (with expected results)

1. **(Must)** All values of $\log(-1)$ and $\operatorname{Log}(-1)$. *Expected:* $i\pi(2k+1)$; the principal value is $i\pi$.
2. **(Must)** $\operatorname{Log}(1+i)$. *Expected:* $\frac{1}{2}\ln 2 + i\pi/4$.
3. **(Must)** $i^i$ (principal). *Expected:* $e^{-\pi/2} \approx 0.2079$, a real number.
4. **(Must)** Solve $\sin z = 2$. *Expected:* $z = \frac{\pi}{2} + 2\pi k \pm i\ln(2+\sqrt{3})$.
5. **(Must)** Show $|\sin z|^2 = \sin^2 x + \sinh^2 y$, and conclude $\sin z$ is unbounded. *Expected:* expand $\sin(x+iy)$ and take the modulus.
6. **(Should)** Both values of $(1+i)^{1/2}$. *Expected:* $\pm 2^{1/4} e^{i\pi/8}$.
7. **(Should)** Show $\operatorname{Log}(z_1 z_2) \neq \operatorname{Log} z_1 + \operatorname{Log} z_2$ in general. *Expected:* for $z_1 = z_2 = -1$, the left side is $\operatorname{Log} 1 = 0$ and the right side is $2\pi i$.
8. **(Should)** Find the branch points of $\sqrt{z^2-1}$ and a cut that makes it single-valued. *Expected:* branch points $\pm 1$; the cut along $[-1,1]$ gives a function analytic outside the segment with $f \sim z$ at infinity (a large loop makes $z^2-1$ wind twice, so the square root returns to itself).
9. **(Could)** $\frac{d}{dz} z^{1/2}$ at $z = 4$ on the principal branch. *Expected:* $\frac{1}{4}$.
10. **(Could)** Show $\cos^{-1} z = -i \log\left(z + i\sqrt{1-z^2}\right)$ and evaluate $\cos^{-1} 2$. *Expected:* $\pm i\ln(2+\sqrt{3})$ plus multiples of $2\pi$.

## Optional extension

Parametrise a small circle around $z = 0$ and follow $\sqrt{z}$ once around: the value changes sign. Do the same for $\log z$ (it gains $2\pi i$). That is the definition of a branch point, done by hand.

## Pitfalls

- Writing $\log(z_1 z_2) = \log z_1 + \log z_2$ without tracking the $2\pi i$.
- Using the principal branch where the problem needs another.
- Treating $\sin z$ as bounded by 1.
- Calling the cut a property of the function. The cut is a choice; the branch point is not.

## Gate questions

1. Why must $\log z$ be multivalued?
2. What is the difference between a branch point and a branch cut?
3. Where is the principal logarithm analytic, and why not on the negative real axis?

## One-page cheat sheet contents

$\log z$ and $\operatorname{Log} z$; $z^a$; the $\sin z = w$ recipe; the branch-point test (follow a loop); the three counterexamples.

## Where this is used

CA06 (the Laurent series of log is not available at a branch point), Part 2 (keyhole contours, Riemann surfaces), QFT thresholds.

# CA04: Contour integrals and Cauchy's theorem

**Books:** Ávila Chapter 3, pp. 75 to 101 (exercise sets on pp. 79, 86, 99; answers and hints on pp. 88, 101). Churchill Chapter 4. Stewart 16.2 to 16.4 as the real-variable parent. **Time:** 4 sessions (Must: 4). **Tier note:** no skip-test passes this module by memory. The proof of Cauchy's theorem from Green's theorem is the point, so do it by hand even if you know the statement.

## Why this matters

A contour integral is a line integral of the type $\int P\,dx + Q\,dy$ with complex $P$ and $Q$. Everything you did in Stewart 16.2 to 16.4 transfers: parametrise the curve, substitute, integrate, with orientation mattering. Cauchy's theorem is Green's theorem plus the Cauchy–Riemann equations, so it is a corollary of what you already own. The deformation principle, which you met as the extended Green's theorem with a hole, becomes the working method of the whole guide: shrink a contour onto small circles around the singularities. The ML-estimate is the tool that will make Jordan's lemma work in CA08. The wire field you studied (curl-free, but circulation $2\pi$) is $1/z$ in disguise.

## Skip-test (40 minutes, no notes)

1. Write $\int_C f\,dz$ as two real line integrals in $u$ and $v$.
2. State and prove the ML-estimate.
3. Compute $\oint_{|z|=1} z^n\,dz$ for every integer $n$.
4. Prove Cauchy's theorem from Green's theorem.
5. Explain why $\oint_{|z|=1} dz/z \neq 0$ does not contradict Cauchy's theorem.
6. State the deformation principle for a region with one hole.

## Prerequisites

CA02 (Cauchy–Riemann). Stewart 16.2 (line integrals), 16.3 (path independence, primitives), 16.4 (Green's theorem and its extended version).

## Detailed topics (checklist)

**[Must]** Arcs and contours: parametrised, piecewise smooth (regular), simple, closed; positive orientation. Name the Jordan curve theorem (a simple closed curve has an inside and an outside).

**[Must]** Definition and evaluation: $\int_C f\,dz = \int_a^b f(\gamma(t))\gamma'(t)\,dt$. Equals $\int u\,dx - v\,dy + i\int v\,dx + u\,dy$. Independent of the parametrisation, changes sign when the orientation reverses, additive on the pieces of a contour.

**[Must]** The ML-estimate: $\left|\int_C f\,dz\right| \leq ML$, where $M = \max_C|f|$ and $L$ is the length of $C$.

**[Must]** Primitives: if $F' = f$ on a region containing $C$, then $\int_C f\,dz = F(b) - F(a)$ (Stewart 16.3), so closed integrals vanish.

**[Must]** Cauchy's theorem: if $f$ is analytic in a simply connected domain, $\oint_C f\,dz = 0$ for every closed contour in it. Ávila proves it from Green's theorem (p. 89), assuming a continuous $f'$; Goursat's theorem removes the assumption (name it).

**[Must]** Deformation. In a multiply connected region, two closed contours that can be deformed into each other without crossing a singularity give the same integral. The master integral:

$$\oint_{|z-z_0|=r} (z-z_0)^n\,dz = 2\pi i\,\delta_{n,-1}.$$

**[Should]** A primitive exists in a simply connected domain; Morera's theorem (the converse of Cauchy), by name.

## Method cards

**Evaluating a contour integral directly.** (1) Parametrise: $z = \gamma(t)$, $t \in [a, b]$. (2) Write endpoints $\gamma(a)$, $\gamma(b)$ and check them, as in your Stewart 16.2 work. (3) Compute $\gamma'(t)$. (4) Substitute into $f(\gamma(t))\gamma'(t)$. (5) Integrate. Direction lives in the parametrisation only.

**ML-estimate.** (1) Bound $|f|$ on the contour by a number $M$ (find the minimum of the denominator). (2) Compute the length $L$. (3) Conclude $|\int| \le ML$. Use it to show an arc contributes nothing as its radius grows.

**Deformation.** (1) Mark the singularities. (2) Decide which lie inside the contour. (3) Replace the contour by small circles around those singularities (a channel between them cancels). (4) Evaluate each circle with the master integral.

## Reading plan

| Session | Task |
|---|---|
| A1 | Ávila pp. 75 to 82 (arcs, Jordan's theorem, contour integral, invariance, properties). Derive items 1 and 3 |
| B1 | Exercises 1 to 3 and the Ávila sets on pp. 79 and 86 |
| A2 | Ávila pp. 89 to 93 (Green's theorem, Cauchy's theorem). Derive item 2, the proof by hand |
| B2 | Exercises 4 to 6 |
| A3 | Ávila pp. 93 to 101 (contour integrals and primitives, deformation) |
| B3 | Exercises 7 to 10 and the Ávila set on p. 99 |
| A4 | Spare session: redo the failed items of the skip-test. Cheat sheet |

## Derive from scratch

1. $\oint_C (u+iv)(dx+i\,dy) = \oint_C (u\,dx - v\,dy) + i\oint_C (v\,dx + u\,dy)$, and both real integrals vanish by Green's theorem and Cauchy–Riemann. For the first, $P=u$ and $Q=-v$, so $Q_x - P_y = -v_x - u_y = 0$. For the second, $P=v$ and $Q=u$, so $Q_x - P_y = u_x - v_y = 0$.
2. The master integral: with $z = z_0 + re^{i\theta}$, $\oint (z-z_0)^n\,dz = ir^{n+1}\int_0^{2\pi} e^{i(n+1)\theta}\,d\theta = 2\pi i\,\delta_{n,-1}$.
3. The ML-estimate, from $\left|\int_a^b g\,dt\right| \le \int_a^b |g|\,dt$.

## Exercises (with expected results)

1. **(Must)** $\oint_{|z|=1} dz/z$ and $\oint_{|z|=1} z^2\,dz$. *Expected:* $2\pi i$ and 0.
2. **(Must)** Compute $\int_C \bar{z}\,dz$ along the upper half of the unit circle from 1 to $-1$, then along the straight segment. *Expected:* $i\pi$ and 0 (so it is path-dependent, and $\bar{z}$ is not analytic).
3. **(Must)** $\int_0^{1+i} z^2\,dz$ along any path. *Expected:* $\frac{(1+i)^3}{3} = \frac{-2+2i}{3}$.
4. **(Must)** Bound $\left|\oint_{|z|=2} \frac{dz}{z^2+1}\right|$ with the ML-estimate, then compute it exactly with partial fractions. *Expected:* the bound is $\frac{4\pi}{3}$ (since $|z^2+1| \ge 3$ on the circle); the exact value is 0.
5. **(Must)** Prove Cauchy's theorem by hand, by the expansion in derivation item 1. *Expected:* both double integrals vanish identically.
6. **(Should)** $\oint \frac{dz}{z(z-1)}$ on $|z| = \frac{1}{2}$ and on $|z| = 2$. *Expected:* $-2\pi i$ and 0. Use $\frac{1}{z(z-1)} = \frac{1}{z-1} - \frac{1}{z}$. This is "which singularities you enclose".
7. **(Should)** For the positively oriented square with vertices $\pm 2 \pm 2i$, compute $\oint \frac{dz}{z-z_0}$ for $z_0 = \frac{1}{2}$ and $z_0 = 3$. *Expected:* $2\pi i$ and 0, by deforming to a circle.
8. **(Should)** $\int_0^{i\pi} e^z\,dz$. *Expected:* $-2$.
9. **(Could)** Show $1/z$ has no primitive on $\mathbb{C}\setminus\{0\}$ but has one on the slit plane. *Expected:* a primitive would make $\oint dz/z = 0$; on the slit plane it is $\operatorname{Log} z$.
10. **(Could)** The wire field. For $\mathbf{F} = \dfrac{-y\,\mathbf{i} + x\,\mathbf{j}}{x^2+y^2}$ show $\oint \mathbf{F}\cdot d\mathbf{r} = \operatorname{Re}\oint (P - iQ)\,dz$ and that $P - iQ = -i/z$. *Expected:* $2\pi$ on any loop around the origin.

## Optional extension

Show Cauchy's theorem for a region with a hole by cutting a channel (your Stewart 16.4 Example 5, now in complex form). State what changes if the integrand has two singularities inside.

## Pitfalls

- Dropping the orientation: a clockwise circle gives $-2\pi i$.
- Applying Cauchy's theorem when a singularity sits inside or on the contour.
- Forgetting the $\gamma'(t)$ factor.
- Bounding $|f|$ at one point instead of its maximum (for the ML-estimate use the minimum of the denominator).

## Gate questions

1. Why is Cauchy's theorem a corollary of Green's theorem?
2. Why is $\oint dz/z = 2\pi i$ not a contradiction?
3. Why is the integral over a contour independent of its shape as long as it avoids the singularities?

## One-page cheat sheet contents

The two-real-integrals formula; the parametrisation recipe; the ML-estimate; Cauchy's theorem with its hypotheses; the master integral; the deformation recipe.

## Where this is used

CA05 (Cauchy's formula), CA07 and CA08 (every residue calculation), Jackson Chapter 1 (Green's identities in the same spirit).

# CA05: Cauchy's integral formula, maximum modulus and harmonic functions

**Books:** Ávila Chapter 3, pp. 101 to 117 (exercise sets on pp. 107 and 116; answers on pp. 109 and 117). Churchill Chapter 4. **Time:** 5 sessions (Must: 2, Should: 1, Could: 2). **Tier note:** the Could part (Dirichlet and Neumann, pp. 114 to 117) reads better after EM Modules 04 and 05, as a second proof of uniqueness. Do not do it first.

## Why this matters

Cauchy's formula says the values of an analytic function on a closed contour determine it everywhere inside. Nothing in real analysis behaves like this. The consequences are the whole guide: analytic functions are infinitely differentiable, they equal their Taylor series, and a residue at a simple pole is just Cauchy's formula read backwards. The mean-value property and the maximum modulus principle give "harmonic functions have no interior maxima or minima", which is Earnshaw's theorem for the electrostatic potential in a charge-free region and the mean-value theorem in Jackson's first chapter. The Dirichlet problem, boundary values determining a harmonic function, is the 2D version of what Jackson proves in Chapter 1.

## Skip-test (40 minutes, no notes)

1. State Cauchy's integral formula and its derivative version.
2. Evaluate $\oint_{|z|=2} \frac{e^z}{z-1}\,dz$.
3. State the mean-value property for an analytic function.
4. State the maximum modulus principle.
5. Explain why a harmonic function has no interior maximum.

## Prerequisites

CA04.

## Detailed topics (checklist)

**[Must]** Cauchy's integral formula: $f(z_0) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z-z_0}\,dz$ for $z_0$ inside $C$. Derivatives of all orders: $f^{(n)}(z_0) = \frac{n!}{2\pi i}\oint_C \frac{f(z)}{(z-z_0)^{n+1}}\,dz$. Evaluation of contour integrals with one or several singularities by the formula and partial fractions.

**[Must]** Consequence: analytic implies infinitely differentiable (this removes the $C^2$ assumption of CA02).

**[Should]** The Cauchy estimates $|f^{(n)}(z_0)| \le n!M/R^n$ and Liouville's theorem (a bounded entire function is constant). The mean-value property $f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + re^{i\theta})\,d\theta$. The maximum modulus principle. A harmonic function has no interior extremum.

**[Could]** Harmonic functions and conjugates; a harmonic function determines the analytic function up to a constant; multiply connected regions; the Dirichlet and Neumann problems (Ávila pp. 109 to 117).

## Method cards

**Cauchy's formula as a recipe.** (1) Identify $f(z)$, the part analytic inside, and $z_0$. (2) Check $z_0$ is inside the contour. (3) The integral is $2\pi i\, f(z_0)$. With a higher power $(z-z_0)^{n+1}$ in the denominator, the integral is $2\pi i\, f^{(n)}(z_0)/n!$.

**Several singularities.** Use partial fractions and apply the formula to each term, or deform to small circles.

## Reading plan

| Session | Task |
|---|---|
| A1 | Ávila pp. 101 to 109 (Cauchy's formula, derivatives of all orders). Derive items 1 and 2 |
| B1 | Exercises 1 to 5 and the Ávila set on p. 107 |

| Session | Task |
|---|---|
| A2 | Ávila pp. 109 to 113 (harmonic functions, maximum modulus principle). Derive items 3 and 4 [Should] |
| A3 | Ávila pp. 114 to 117 (Dirichlet and Neumann), after EM Module 05 [Could] |
| B3 | Exercises 9 and 10 and the Ávila set on p. 116 [Could] |

(Exercises 6 to 8, Should, go with A2.)

## Derive from scratch

1. Cauchy's formula from the master integral: shrink the contour onto a small circle around $z_0$, where $f(z) \approx f(z_0)$.
2. The derivative formula, by differentiating under the integral sign.
3. The mean-value property, from the formula on a circle: $z = z_0 + re^{i\theta}$ gives $dz/(z - z_0) = i\,d\theta$.
4. The maximum modulus principle, from the mean-value property.

## Exercises (with expected results)

1. **(Must)** $\oint_{|z|=2} \frac{e^z}{z-1}\,dz$. *Expected:* $2\pi ie \approx 17.08\,i$.
2. **(Must)** $\oint_{|z|=2} \frac{\cos z}{z^3}\,dz$. *Expected:* $-\pi i$ (derivative version with $n = 2$ and $f''(0) = -1$).
3. **(Must)** $\oint_{|z|=3} \frac{z^2}{(z-1)(z-2)}\,dz$. *Expected:* $6\pi i$. Hint: $\frac{z^2}{(z-1)(z-2)} = 1 + \frac{4}{z-2} - \frac{1}{z-1}$.
4. **(Must)** $\oint_{|z|=1} \frac{\sin z}{z^2}\,dz$. *Expected:* $2\pi i$.
5. **(Must)** Verify the mean-value property for $f = z^2$ on the circle $|z - 1| = 0.7$. *Expected:* the average is $f(1) = 1$.
6. **(Should)** Outline the proof of Liouville's theorem from $|f'(z_0)| \le M/R$. *Expected:* let $R \to \infty$, so $f' \equiv 0$.
7. **(Should)** Find the maximum of $|z^2 + 1|$ on the closed unit disk. *Expected:* 2, attained at $z = \pm 1$ on the boundary (the interior value at 0 is only 1).
8. **(Should)** For $f = z^2$, show the level curves of $u = x^2 - y^2$ and $v = 2xy$ are orthogonal. *Expected:* $\nabla u = (2x, -2y)$ and $\nabla v = (2y, 2x)$ give $\nabla u \cdot \nabla v = 0$.
9. **(Could)** Show a nonconstant harmonic function has no interior local extremum, and say what this means for the electrostatic potential in a charge-free region. *Expected:* every critical point of $V$ there is a saddle (Earnshaw's theorem: no stable static trap).
10. **(Could)** Solve the Dirichlet problem in the unit disk for the boundary values $\cos\theta$ and $\cos^2\theta$. *Expected:* $u = x$ and $u = \frac{1}{2} + \frac{1}{2}(x^2 - y^2)$ (write $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$).

## Optional extension

Prove the maximum modulus principle for $|f|$ from the mean-value property, and then for $u = \operatorname{Re} f$ from the harmonic version. Compare with Jackson's uniqueness proof for the Dirichlet problem.

## Pitfalls

- Applying the formula when $z_0$ lies outside the contour (the integral is then 0).
- Forgetting the factor $n!$ in the derivative version.
- Taking the maximum modulus of an analytic function over an open set; it is attained on the boundary of a compact region.
- Saying "harmonic implies analytic". The correct statement is that a harmonic function is the real part of an analytic function on a simply connected domain.

## Gate questions

1. Why do the values on the boundary determine $f$ inside?
2. Why is an analytic function automatically infinitely differentiable?
3. Why does the mean-value property forbid interior extrema?

## One-page cheat sheet contents

Cauchy's formula and the derivative version; the recipe; the mean-value property; the maximum modulus principle; the Earnshaw statement.

## Where this is used

CA06 and CA07 (Taylor and Laurent coefficients are Cauchy integrals), EM Module 04 (Earnshaw's theorem, uniqueness), Jackson's mean-value problem in Chapter 1.

# CA06: Taylor and Laurent series

**Books:** Ávila Chapter 4, pp. 118 to 150 (exercise sets on pp. 125, 132, 142, 149; hints and answers on pp. 127, 132, 144). Churchill Chapter 5. Stewart Chapter 11 for the real series. **Time:** 6 sessions (Must: 4; the series patch is 1 more session if you need it). **Tier note:** if the series skip-test below fails, do the patch first.

## Why this matters

Laurent is the only genuinely new idea in the series chapter. A Taylor series has non-negative powers; a Laurent series allows negative ones and converges on an annulus, because the centre may be singular. The coefficient of $(z-z_0)^{-1}$ is the residue, so CA07 is a chapter about reading one coefficient. The classification of singularities (removable, pole, essential) is read straight off the negative powers. Taylor's theorem also explains something you saw in Stewart: $\frac{1}{1+x^2}$ is smooth on the whole real line, yet its series about 0 has radius 1, because the singularities sit at $\pm i$ in the complex plane. The binomial series $(1+z)^{-1/2}$ generates the Legendre polynomials, so the multipole expansion of Jackson comes from this chapter.

## Skip-test (40 minutes, no notes)

Series patch, 15 minutes, from Stewart Chapter 11:

1. Find the radius of convergence of $\sum \frac{n!}{n^n} z^n$.
2. Does $\sum \frac{z^n}{n^2}$ converge on $|z| = 1$?
3. Write the Taylor series of $\frac{1}{1+z^2}$ about 0 and give its radius.

Laurent, 25 minutes:

4. State Taylor's theorem for an analytic function and say what limits the radius.
5. Expand $\frac{1}{z(z-1)}$ in $0 < |z| < 1$ and in $|z| > 1$.
6. Write the Laurent coefficient formula.

## Prerequisites

CA04 and CA05. Stewart Chapter 11 (series of constants, convergence tests, power series, radius of convergence).

## Detailed topics (checklist)

**[Must]** Series of functions: pointwise versus uniform convergence; the Weierstrass M-test; a uniformly convergent series of continuous functions has a continuous sum.

**[Must]** Power series: radius of convergence (ratio or root test); termwise differentiation and integration inside the radius.

**[Must]** Taylor's theorem: $f$ analytic in $|z-z_0| < R$ equals its Taylor series there. The radius is the distance to the nearest singularity, which can be invisible on the real axis.

**[Must]** Known series ($e^z$, $\sin z$, $\cos z$, $\frac{1}{1-z}$, $\log(1+z)$, $(1+z)^\alpha$); products and quotients of series.

**[Must]** Laurent series on an annulus $r < |z - z_0| < R$:

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z-z_0)^n, \qquad a_n = \frac{1}{2\pi i} \oint \frac{f(z)}{(z-z_0)^{n+1}}\, dz.$$

The series is unique for a given annulus, and the same function has different Laurent series in different annuli.

**[Should]** Zeros of analytic functions: isolated, with an order; the identity theorem (by name). Behaviour at infinity ("regularidade no infinito"): study $f(1/w)$ at $w = 0$.

**[Could]** The Cauchy–Hadamard formula for the radius.

## Method cards

**Laurent series of a rational function in an annulus.** (1) Partial fractions. (2) For each term $\frac{1}{z-a}$: if $|z| < |a|$ expand $-\frac{1}{a}\sum (z/a)^n$; if $|z| > |a|$ expand $\frac{1}{z}\sum (a/z)^n$. (3) Add. (4) Check the annulus: each

geometric series must converge there.

**Taylor from a known series.** Substitute and multiply. Do not differentiate unless you must.

**Reading the type of a singularity.** No negative powers: removable. Finitely many, down to $(z-z_0)^{-m}$: pole of order $m$. Infinitely many: essential.

## Reading plan

| Session | Task |
|---|---|
| A0 | Series patch, only if the skip-test fails: Stewart 11.9 and 11.10, convergence tests |
| A1 | Ávila pp. 118 to 138 (series of functions, uniform convergence, power series, Taylor, examples, products). Derive items 1 and 2 |
| B1 | Exercises 1 and 2 and the Ávila sets on pp. 125, 132 and 142 |
| A2 | Ávila pp. 144 to 150 (Laurent, infinity, zeros). Derive item 3 |
| B2 | Exercises 3 to 5 and the Ávila set on p. 149 |
| B3 | Should: exercises 6 to 10 and the cheat sheet |

## Derive from scratch

1. Taylor's theorem from Cauchy's formula: expand $\frac{1}{z-\zeta}$ as a geometric series inside the circle.
2. The radius of convergence of the Taylor series of $\frac{1}{1+z^2}$ at 0, and why it is 1.
3. The Laurent coefficient formula, from the formula on the two circles bounding the annulus.

## Exercises (with expected results)

1. **(Must)** Radius of convergence of $\sum_{n\geq 1} \frac{n!}{n^n} z^n$. *Expected:* $e$.
2. **(Must)** Taylor series of $\frac{1}{1+z^2}$ about 0. *Expected:* $\sum (-1)^n z^{2n}$ with radius 1; the singularities are at $\pm i$, invisible from the real line.
3. **(Must)** Laurent series of $\frac{1}{z(z-1)}$ in $0<|z|<1$ and in $|z|>1$. *Expected:* $-\frac{1}{z}-1-z-z^2-\cdots$ and $z^{-2}+z^{-3}+\cdots$.
4. **(Must)** Laurent series of $\frac{1}{(z-1)(z-2)}$ in $1<|z|<2$. *Expected:* $-\sum_{n\geq 0}\frac{z^n}{2^{n+1}}-\sum_{n\geq 1} z^{-n}$.
5. **(Must)** Show $e^{1/z}=\sum_{n\geq 0}\frac{1}{n!\,z^n}$ has an essential singularity at 0 and residue 1. *Expected:* infinitely many negative powers, $a_{-1}=1$.
6. **(Should)** The coefficient $a_{-1}$ of $\frac{e^z}{z^3}$ at 0. *Expected:* $\frac{1}{2}$.
7. **(Should)** The Laurent series of $\cot z$ about 0. *Expected:* $\frac{1}{z}-\frac{z}{3}-\frac{z^3}{45}-\cdots$ (residue 1).
8. **(Should)** The order of the zero at 0 of $z-\sin z$ and of $1-\cos z$. *Expected:* 3 and 2.
9. **(Could)** Check the coefficient formula by computing $a_{-1}$ of $\frac{1}{z(z-1)}$ at 0 as $\frac{1}{2\pi i}\oint_{|z|=1/2}$. *Expected:* $-1$, matching exercise 3.
10. **(Could)** Expand $(1-2xt+t^2)^{-1/2}$ in powers of $t$ to second order. *Expected:* $1+xt+\frac{3x^2-1}{2}t^2$, i.e. $P_0=1$, $P_1=x$, $P_2=\frac{3x^2-1}{2}$.

## Optional extension

Expand $\frac{1}{z^2-3z+2}$ in all three regions $|z|<1$, $1<|z|<2$, $|z|>2$ and check the three series against each other on the overlaps of their coefficient formulas.

## Pitfalls

- Using the Taylor series outside its disk.
- Using one Laurent series for a whole region when the function has several annuli.
- Computing the radius from the real line only.
- Confusing the Laurent series of $\frac{1}{z-a}$ in $|z|<|a|$ with the one in $|z|>|a|$.

## Gate questions

1. Why does the Taylor series of $\frac{1}{1+x^2}$ have radius 1?
2. Why does a Laurent series converge on an annulus and not on a disk?
3. How do you read the type of a singularity from the series?

## One-page cheat sheet contents

Taylor's theorem and the radius rule; the Laurent formula; the annulus recipe; the singularity types; the standard series.

## Where this is used

CA07 (residues are $a_{-1}$), EM Modules 06 and 07 (the Legendre generating function), the multipole expansion in Jackson Chapter 4.

# CA07: Singularities and the residue theorem

**Books:** Ávila Chapter 5, pp. 151 to 161 (exercise sets on pp. 156 and 160; answers on pp. 157 and 161). Churchill Chapter 6. **Time:** 4 sessions (Must: 4). **Tier note:** this module is the spine of the guide. Do not skip it, and do all of exercises 1 to 5.

## Why this matters

A residue is the coefficient $a_{-1}$ of a Laurent series, and the residue theorem turns a contour integral into a sum of residues. Everything in CA08 and CA09 is this theorem applied to a particular integrand. The classification of singularities is also physical: in QFT a pole of a propagator is a stable particle and a branch cut is a multi-particle threshold, and the distinction begins here. The retarded, advanced and Feynman Green functions come from simple poles, so simple poles are your working case, but you need the formulas for higher orders as well.

## Skip-test (40 minutes, no notes)

1. Classify the singularities of $\frac{\sin z}{z}$, $\frac{1}{z^2(z-1)}$, $e^{1/z}$ and $\frac{z}{\sin z}$.
2. Write the residue formula for a simple pole and for a pole of order $m$.
3. Find the residue of $\frac{e^z}{(z-1)^3}$ at $z = 1$.
4. State the residue theorem and its hypotheses.
5. Compute $\oint_{|z|=3} \frac{z+1}{z^2-2z}\,dz$.

## Prerequisites

CA04, CA05, CA06.

## Detailed topics (checklist)

[**Must**] Isolated singularities and their types: removable, pole of order $m$, essential (Ávila pp. 151 to 154). Read the type from the limit of $(z-z_0)^m f(z)$ or from the Laurent series. Name the Casorati–Weierstrass theorem.

[**Must**] The residue $\mathrm{Res}(f, z_0) = a_{-1}$.

[**Must**] The residue theorem: if $C$ is a positively oriented simple closed contour and $f$ is analytic on and inside $C$ except at finitely many points $z_k$ inside,

$$\oint_C f(z)\,dz = 2\pi i \sum_k \mathrm{Res}(f, z_k).$$

[**Must**] Computing residues: simple pole, $g/h$ with a simple zero of $h$, higher order, and by the Laurent series. At an essential singularity you must expand.

[**Should**] If $f = O(z^{-2})$ at infinity and all singularities are inside $C$, the sum of all residues is 0.

## Method cards

**Residues.**

- Simple pole: $\mathrm{Res} = \lim_{z\to z_0}(z-z_0)f(z)$.
- $f = g/h$ with $h(z_0) = 0$ and $h'(z_0) \neq 0$: $\mathrm{Res} = g(z_0)/h'(z_0)$.
- Pole of order $m$: $\mathrm{Res} = \dfrac{1}{(m-1)!}\lim_{z\to z_0}\dfrac{d^{m-1}}{dz^{m-1}}\big[(z-z_0)^m f(z)\big]$.
- Essential singularity (or anything unclear): expand in a Laurent series and read $a_{-1}$.

**Using the theorem.** (1) Draw the contour and mark the singularities. (2) Decide which are inside; poles on the contour need CA08. (3) Compute each residue. (4) Add, multiply by $2\pi i$. (5) Check the orientation: clockwise gives $-2\pi i$.

## Reading plan

| Session | Task |
|---|---|
| A1 | Ávila pp. 151 to 157 (isolated singularities, the three types). Derive item 1 |
| B1 | Exercises 1 to 3 and the Ávila set on p. 156 |
| A2 | Ávila pp. 157 to 161 (the residue theorem). Derive items 2 and 3 |
| B2 | Exercises 4 to 10 and the Ávila set on p. 160. Cheat sheet |

## Derive from scratch

1. The three types of singularity from the Laurent series.
2. The residue theorem: deform to small circles (CA04) and apply the master integral to each Laurent series term.
3. The formula for the residue at a pole of order $m$.

## Exercises (with expected results)

1. **(Must)** Classify the singularities of $\frac{\sin z}{z}$, $\frac{1}{z^2(z-1)}$, $e^{1/z}$ and $\frac{z}{\sin z}$. *Expected:* removable at 0; a double pole at 0 and a simple pole at 1; essential at 0; removable at 0 and simple poles at $n\pi$, $n \neq 0$.
2. **(Must)** The residues of $\frac{1}{z^2+1}$. *Expected:* $-\frac{i}{2}$ at $z = i$ and $+\frac{i}{2}$ at $z = -i$.
3. **(Must)** The residue of $\frac{e^z}{z^2}$ at 0 and of $\frac{e^z}{(z-1)^3}$ at 1. *Expected:* 1 and $\frac{e}{2}$.
4. **(Must)** $\oint_{|z|=3} \frac{z+1}{z^2-2z}\,dz$. *Expected:* $2\pi i$ (residues $-\frac{1}{2}$ at 0 and $\frac{3}{2}$ at 2).
5. **(Must)** $\oint_{|z|=1} \frac{\sin z}{z^4}\,dz$. *Expected:* $-\frac{i\pi}{3}$ (the residue is the coefficient of $z^3$ in $\sin z$, i.e. $-\frac{1}{6}$).
6. **(Should)** $\oint \cot z\,dz$ on $|z| = 1$ and on $|z| = 4$. *Expected:* $2\pi i$ and $6\pi i$ (residue 1 at each $n\pi$; the circle of radius 4 encloses 0 and $\pm\pi$).
7. **(Should)** $\oint_{|z|=1} \sin(1/z)\,dz$. *Expected:* $2\pi i$ (essential singularity; the residue is 1).
8. **(Should)** $\oint_{|z|=1.5} \frac{dz}{z^4+1}$. *Expected:* 0, since all four poles are inside and $f = O(z^{-4})$, so the residues sum to zero.
9. **(Could)** The residue of $\frac{1}{(z^2+1)^2}$ at $z = i$, and $\oint_{|z-i|=1} \frac{dz}{(z^2+1)^2}$. *Expected:* $-\frac{i}{4}$ and $\frac{\pi}{2}$.
10. **(Could)** Locate the poles of $\frac{1}{k^2-m^2+i\epsilon}$ and of $\frac{1}{k^2-m^2-i\epsilon}$. *Expected:* for $+i\epsilon$, at $k = \pm(m - i\epsilon')$, i.e. $+m - i\epsilon'$ below the real axis and $-m + i\epsilon'$ above; for $-i\epsilon$ the two poles swap sides. (This is EM Module 02, exercise 7.)

## Optional extension

Prove that the sum of the residues of a rational function that decays like $z^{-2}$ is zero, using a large circle, and use it to get the answer to exercise 8 without computing a single residue.

## Pitfalls

- Using the simple-pole formula on a pole of higher order.
- Forgetting that a pole on the contour is not covered by the theorem (CA08).
- Including a singularity that lies outside the contour.
- Missing the factor $\frac{1}{(m-1)!}$.
- Forgetting that an essential singularity needs the full Laurent expansion.

## Gate questions

1. Why does only $a_{-1}$ survive in $\oint f\,dz$?
2. How do you read the order of a pole from the Laurent series?
3. Why is the sum of residues of a rational function $O(z^{-2})$ equal to zero?

## One-page cheat sheet contents

The three types and how to read them; the four residue formulas; the residue theorem with its hypotheses; the orientation rule; the "sum of residues is zero" shortcut.

### Where this is used

CA08 and CA09 (every closed contour), EM Modules 02, 11 and 19, and the pole-as-particle picture in QFT.

# CA08: Improper integrals, Jordan's lemma and principal values

**Books:** Ávila Chapter 5, pp. 161 to 169 (rational integrands, Jordan's lemma) and pp. 173 to 174 (trigonometric integrals). Exercise sets on pp. 163, 168 and 174; answers on pp. 164 and 169. **Churchill Chapter 7** for indented contours and the Cauchy principal value: **Ávila's table of contents has no section on them, so Churchill (or Boas) carries that part. Time:** 4 sessions (Must: 4). **Tier note:** the three-star module. Do all of exercises 1 to 5, and do not leave it before exercise 6 is clear.

## Why this matters

Jackson's Green function for the wave equation is a Fourier integral whose denominator vanishes on the real axis. It is undefined until you choose how to go around the poles. The calculation is exactly this module: close the real line with a large semicircle, justify the closure with Jordan's lemma, and collect residues. Indenting the contour around a pole on the real axis, and the principal value that results, is what makes retarded, advanced and Feynman different, and it is the tool behind the Kramers–Kronig relations in EM Module 19. In your own words: the indentation direction is the physical boundary condition.

## Skip-test (40 minutes, no notes)

1. Evaluate $\int_{-\infty}^{\infty} \frac{dx}{x^2+1}$ by closing the contour, and justify discarding the arc.
2. State Jordan's lemma.
3. Evaluate $\int_{-\infty}^{\infty} \frac{\cos kx}{x^2+a^2}\,dx$ for $k, a > 0$.
4. Explain which half-plane to close in for $e^{iax}$ with $a > 0$ and with $a < 0$.
5. Evaluate $\int_{-\infty}^{\infty} \frac{\sin x}{x}\,dx$ with an indented contour.
6. Write $\frac{1}{x \pm i\epsilon}$ in terms of the principal value and a delta function.

## Prerequisites

CA04 (the ML-estimate), CA07 (residues). Stewart 7.8 (improper integrals, convergence at both ends).

## Detailed topics (checklist)

**[Must]** Rational integrands on the real line: close with a large semicircle $C_R$; show $\int_{C_R} \to 0$ by the ML-estimate when the degree of the denominator exceeds that of the numerator by at least 2.

**[Must]** Jordan's lemma: if $g$ is continuous in the upper half-plane for $|z| \geq R_0$ and $\max_{C_R}|g| \to 0$, then for $a > 0$, $\int_{C_R} g(z)e^{iaz}\,dz \to 0$. For $a < 0$ close in the lower half-plane. The proof uses $\sin\theta \geq 2\theta/\pi$ on $[0, \pi/2]$ to get $\int_0^{\pi} e^{-aR\sin\theta}\,d\theta < \pi/(aR)$.

**[Must]** Trigonometric integrals $\int_0^{2\pi} R(\cos\theta, \sin\theta)\,d\theta$: put $z = e^{i\theta}$, $\cos\theta = \frac{1}{2}(z + z^{-1})$, $\sin\theta = \frac{1}{2i}(z - z^{-1})$, $d\theta = dz/(iz)$, and integrate over the unit circle (Ávila pp. 173 to 174).

**[Must]** Poles on the real axis: indent the contour with a small semicircle. A small semicircle of radius $\rho$ around a simple pole, traversed clockwise (passing above it), contributes $-i\pi\,\mathrm{Res}$ as $\rho \to 0$. The result is the Cauchy principal value (Churchill Chapter 7).

**[Should]** The Sokhotski formula $\frac{1}{x \pm i\epsilon} = \mathrm{PV}\frac{1}{x} \mp i\pi\delta(x)$ and its use with the $\pm i\epsilon$ prescription.

**[Could]** The multivalued integrands of Ávila pp. 169 to 173 (keyhole contours) are deferred to Part 2 (appendix).

## Method cards

**Closing a contour.** (1) Write the integrand, often $e^{iax}g(x)$. (2) Decide the half-plane: $a > 0$ upward, $a < 0$ downward. For a rational integrand with no exponential, either, by the ML-estimate. (3) Find the poles in that half-plane; $\pm i\epsilon$ tells you which side of the real axis each real pole moves to. (4) Sum $2\pi i\times$ residues (counterclockwise, upward) or $-2\pi i\times$ residues (clockwise, downward). (5) Check the limit $x \to 0$ and any symmetry.

**Indenting.** (1) Take the principal value outside small neighbourhoods of the real poles. (2) Close with the big semicircle. (3) Add a small semicircle above each real pole (pole outside the contour). (4) Then $\mathrm{PV}\int = 2\pi i \sum_{\text{inside}} \mathrm{Res} + i\pi \sum_{\text{on axis}} \mathrm{Res}$ when closing upward.

**Trigonometric integral.** (1) Substitute $z = e^{i\theta}$. (2) Find the poles inside the unit circle. (3) Residues, times $2\pi i$.

## Reading plan

| Session | Task |
|---|---|
| A1 | Ávila pp. 161 to 168 (rational integrands, Jordan's lemma). Derive items 1 and 2 |
| B1 | Exercises 1 to 3 and the Ávila sets on pp. 163 and 168 |
| A2 | Ávila pp. 173 to 174 (trigonometric integrals), then Churchill Chapter 7 (indented paths, principal value). Derive items 3 and 4 |
| B2 | Exercises 4 to 10 and the Ávila set on p. 174. Cheat sheet |

## Derive from scratch

1. $\int_{-\infty}^{\infty} \frac{dx}{x^2+1} = \pi$, including the arc estimate.
2. Jordan's lemma, from $\sin\theta \geq 2\theta/\pi$.
3. The indentation lemma: a small clockwise semicircle around a simple pole gives $-i\pi\,\mathrm{Res}$.
4. $\int_{-\infty}^{\infty} \frac{\sin x}{x}\,dx = \pi$ from $\mathrm{PV}\int \frac{e^{ix}}{x}\,dx = i\pi$.

## Exercises (with expected results)

1. **(Must)** $\int_{-\infty}^{\infty} \frac{dx}{x^2+1}$. *Expected:* $\pi$.
2. **(Must)** $\int_{-\infty}^{\infty} \frac{\cos kx}{x^2+a^2}\,dx$ for $k, a > 0$. *Expected:* $\frac{\pi}{a}e^{-ka}$. (Use $e^{ikx}$ and Jordan's lemma.)
3. **(Must)** $\int_{0}^{\infty} \frac{dx}{x^4+1}$. *Expected:* $\frac{\pi}{2\sqrt{2}} \approx 1.1107$ (the whole-line integral is $\frac{\pi}{\sqrt{2}}$; the integrand is even).
4. **(Must)** $\int_0^{2\pi} \frac{d\theta}{2+\cos\theta}$. *Expected:* $\frac{2\pi}{\sqrt{3}}$. Hint: the integral becomes $\oint \frac{2\,dz}{i(z^2+4z+1)}$ with the pole $-2+\sqrt{3}$ inside.
5. **(Must)** $\int_{-\infty}^{\infty} \frac{\sin x}{x}\,dx$, by the indented contour. *Expected:* $\pi$ (from $\mathrm{PV}\int \frac{e^{ix}}{x}\,dx = i\pi$ and taking the imaginary part).
6. **(Should)** $\int_{-\infty}^{\infty} \frac{e^{ikx}}{k^2-m^2\pm i\epsilon}\,dk$, closing for $x > 0$ and $x < 0$. *Expected:* for $+i\epsilon$, $-\frac{i\pi}{m}e^{-im|x|}$; for $-i\epsilon$, $+\frac{i\pi}{m}e^{+im|x|}$. (EM Module 02, exercise 7.)
7. **(Should)** $\int_{-\infty}^{\infty} \frac{x\sin kx}{x^2+a^2}\,dx$ for $k, a > 0$. *Expected:* $\pi e^{-ka}$.
8. **(Should)** For the damped oscillator response $\chi(\omega) = \frac{1}{\omega_0^2-\omega^2-i\gamma\omega}$ (with $e^{-i\omega t}$ time dependence): find its poles and show they lie in the lower half-plane, so $\chi$ is analytic in the upper half-plane; then verify numerically at one frequency that $\mathrm{Re}\,\chi(\omega) = \frac{1}{\pi}\mathrm{PV}\int \frac{\mathrm{Im}\,\chi(\omega')}{\omega'-\omega}\,d\omega'$. *Expected:* poles at $\omega = \frac{-i\gamma \pm \sqrt{4\omega_0^2-\gamma^2}}{2}$; for $\omega_0 = 1$, $\gamma = 0.5$, $\omega = 0.7$, $\chi \approx 1.3330 + 0.9148\,i$ and both sides of the relation agree.
9. **(Could)** $\mathrm{PV}\int_{-\infty}^{\infty} \frac{\cos x}{x^2-1}\,dx$. *Expected:* $-\pi\sin 1 \approx -2.6436$. Hint: poles at $\pm 1$ on the axis with residues $\frac{e^{i}}{2}$ and $-\frac{e^{-i}}{2}$.
10. **(Could)** $\int_{-\infty}^{\infty} \frac{dx}{(x^2+1)^2}$. *Expected:* $\frac{\pi}{2}$ (a double pole at $i$ with residue $-\frac{i}{4}$).

## Optional extension

Do exercise 6 with finite $\epsilon$ and check that as $\epsilon \to 0$ you recover the stated results. Then do exercise 5 again with an indentation below the pole, and show you get the same real integral.

## Pitfalls

- Closing on the wrong side: check where $e^{iax}$ decays.
- Forgetting that real-axis poles make the integral undefined until a prescription is chosen.
- Dropping the arc without checking Jordan's lemma or the ML-estimate.
- Taking the full residue instead of half when the pole is on the contour.
- Getting the sign wrong when closing downward (clockwise).

## Gate questions

1. Why is the choice of how to go around a real-axis pole a physical choice and not a convention?
2. Why does Jordan's lemma work, and what happens for $a < 0$?
3. Why does a small semicircle around a simple pole contribute exactly $\mp i\pi$ Res?

## One-page cheat sheet contents

The closing-contour recipe; Jordan's lemma; the indentation lemma; the principal-value formula; Sokhotski; the trigonometric substitution; the results of exercises 1 to 3, 5 and 6.

## Where this is used

CA09 and EM Module 11 (Green functions of the wave equation), EM Module 19 (Kramers–Kronig), EM Module 02 (exercises 6 to 8).

# CA09: Capstone, the three Green functions, Wick rotation and the gate

**Books:** none new. This module uses your CA07 and CA08 results. Read afterwards: Jackson 6.4 (EM Module 11) and, for the Fourier side, Figueiredo Chapter 6 (linear algebra and Fourier path). **Time:** 2 sessions (Must: 2). **Tier note:** this is the deliverable of the guide. Do not skip it, and do not start it before the CA08 exercises 1 to 5 are clean.

## Why this matters

This is EM Module 11 in miniature. Jackson Fourier-transforms the wave equation and gets $\tilde{G} \propto 1/k^2$ with poles on the real axis. The integral is undefined until you choose a contour, and the three choices give $G_{\text{ret}}$, $G_{\text{adv}}$ and the Feynman propagator. If you can do the one-oscillator version here cold, the field version is a change of notation, and the Klein–Gordon propagator in QFT is the same calculation with $\omega_0^2 \to k^2 + m^2$. The choice of contour is a choice of boundary condition in time: retarded means no response before the source acts; Feynman means positive frequencies propagate forward and negative frequencies backward, which is the origin of antiparticles in the field picture. This module also gives the justification of Wick rotation and the causality-to-analyticity chain behind Kramers–Kronig.

## Skip-test (40 minutes, no notes)

1. Write the equation for the Green function of $\ddot{q} + \omega_0^2 q = \delta(t)$, Fourier transform it, and say why the result is undefined on the real axis.
2. List the three prescriptions, where the poles sit, and the half-plane you close in for each sign of $t$.
3. State which prescription allows Wick rotation, and why.
4. State the Sokhotski formula.
5. Explain why a causal response function is analytic in the upper half-plane.

## Prerequisites

CA07, CA08. Conventions: appendix, section 4 (time dependence $e^{-i\omega t}$, which is the physics convention and the one EM Module 02 uses).

## Detailed topics (checklist)

[**Must**] The oscillator Green function $\ddot{q} + \omega_0^2 q = \delta(t)$. With $q(t) = \int \frac{d\omega}{2\pi} e^{-i\omega t} \tilde{q}(\omega)$, $\tilde{q} = \frac{1}{\omega_0^2 - \omega^2}$, with poles at $\pm\omega_0$ on the real axis.

[**Must**] The three prescriptions:

| Name | Prescription | Poles | Result |
|---|---|---|---|
| Retarded | $\omega \to \omega + i\epsilon$ | both at $\pm\omega_0 - i\epsilon$ (below the axis) | $\theta(t)\,\dfrac{\sin\omega_0 t}{\omega_0}$ |
| Advanced | $\omega \to \omega - i\epsilon$ | both at $\pm\omega_0 + i\epsilon$ (above the axis) | $-\theta(-t)\,\dfrac{\sin\omega_0 t}{\omega_0}$ |
| Feynman | $\omega_0^2 \to \omega_0^2 - i\epsilon$ | $+\omega_0 - i\epsilon'$ below, $-\omega_0 + i\epsilon'$ above | $\dfrac{i}{2\omega_0} e^{-i\omega_0 \lvert t\rvert}$ |

[**Must**] Wick rotation: rotating the contour $\omega \to i\omega_E$ sweeps the first and third quadrants. It is legal only if no pole lies there. The Feynman poles sit in the second and fourth quadrants, so the rotation is allowed; the retarded poles at $\pm\omega_0 - i\epsilon$ include one in the third quadrant, so it is blocked.

[**Should**] The field version: spatial Fourier transform reduces $\Box G = -4\pi\delta^3(\mathbf{x})\delta(t)$ to the oscillator with $\omega_0 = ck$.

[**Should**] Causality and Kramers–Kronig: $\chi(t) = 0$ for $t < 0$ implies $\chi(\omega) = \int_0^\infty \chi(t) e^{i\omega t}\, dt$ is analytic in the upper half-plane, and Cauchy's formula with an indentation gives $\operatorname{Re}\chi = \frac{1}{\pi}\mathrm{PV}\int \frac{\operatorname{Im}\chi(\omega')}{\omega' - \omega}\, d\omega'$ and $\operatorname{Im}\chi = -\frac{1}{\pi}\mathrm{PV}\int \frac{\operatorname{Re}\chi(\omega')}{\omega' - \omega}\, d\omega'$.

[**Could**] The Yukawa potential as a static limit with $\omega_0^2 \to k^2 + \mu^2$.

## Method cards

**The three prescriptions.** (1) Write the denominator with the $\pm i\epsilon$ placed as required. (2) Find the poles and mark which side of the real axis each lies on. (3) For $t>0$ close below (since $e^{-i\omega t}$ decays for $\mathrm{Im}\,\omega<0$), clockwise, so $-2\pi i\sum\mathrm{Res}$; for $t<0$ close above, counterclockwise, $+2\pi i\sum\mathrm{Res}$. (4) If no pole is enclosed the result is 0. (5) Check with the differential equation: the result must solve it away from $t=0$ and have the right jump at $t=0$.

**Residues for this integrand.** Near $\omega=\omega_0$ the denominator behaves like $-2\omega_0(\omega-\omega_0)$, so the residue of $e^{-i\omega t}/(\omega_0^2-\omega^2)$ is $-\frac{e^{-i\omega_0 t}}{2\omega_0}$. Near $\omega=-\omega_0$ it is $+\frac{e^{i\omega_0 t}}{2\omega_0}$.

**Wick rotation.** (1) Locate the poles. (2) Check the first and third quadrants are free of poles. (3) Rotate; the arcs vanish by Jordan's lemma. (4) Continue $t\to -i\tau$ to get the Euclidean propagator.

## Reading plan

| Session | Task |
|---|---|
| A1 | Derive items 1 and 2 below. Exercises 1 to 5 |
| B1 | Exercises 6 to 8, the written page (exercise 10), the gate, the cheat sheet |

## Derive from scratch

1. The retarded, advanced and Feynman functions of the oscillator, with each pole and each half-plane.
2. $G_{\mathrm{ret}}-G_{\mathrm{adv}}=\sin(\omega_0 t)/\omega_0$, and $G_F=\frac{1}{2}(G_{\mathrm{ret}}+G_{\mathrm{adv}})+\frac{i\cos\omega_0 t}{2\omega_0}$.
3. The reason Wick rotation works for $G_F$ and not for $G_{\mathrm{ret}}$.
4. The field version: from $\tilde{G}(k,t)$ to $\delta(t-R/c)/R$ (exercise 8).

## Exercises (with expected results)

1. **(Must)** The three prescriptions for $\int\frac{d\omega}{2\pi}\frac{e^{-i\omega t}}{\omega_0^2-\omega^2}$. *Expected:* retarded $\theta(t)\frac{\sin\omega_0 t}{\omega_0}$; advanced $-\theta(-t)\frac{\sin\omega_0 t}{\omega_0}$; Feynman $\frac{i}{2\omega_0}e^{-i\omega_0|t|}$. (With a finite $\epsilon$ the retarded result carries a factor $e^{-\epsilon t}$.)
2. **(Must)** Show $G_{\mathrm{ret}}-G_{\mathrm{adv}}=\frac{\sin\omega_0 t}{\omega_0}$ for all $t$. *Expected:* the two step functions add to 1.
3. **(Must)** Show $G_F=\frac{1}{2}(G_{\mathrm{ret}}+G_{\mathrm{adv}})+\frac{i}{2\omega_0}\cos\omega_0 t$. *Expected:* $\frac{1}{2}(G_{\mathrm{ret}}+G_{\mathrm{adv}})=\frac{\sin\omega_0|t|}{2\omega_0}$, and $e^{-i\omega_0|t|}=\cos\omega_0 t-i\sin\omega_0|t|$. So the Feynman function is the symmetric part plus a homogeneous solution.
4. **(Must)** Which prescription permits the rotation $\omega=i\omega_E$ without crossing a pole? *Expected:* Feynman (poles in quadrants 2 and 4). The retarded poles include one at $-\omega_0-i\epsilon$ in the third quadrant, so the rotation crosses it.
5. **(Must)** Compute the Euclidean function $\int\frac{d\omega_E}{2\pi}\frac{e^{i\omega_E\tau}}{\omega_0^2+\omega_E^2}$ and relate it to $G_F$. *Expected:* $\frac{e^{-\omega_0|\tau|}}{2\omega_0}$, and $G_F(t\to -i\tau)=i\,G_E(\tau)$.
6. **(Should)** Verify the Sokhotski formula for $f=e^{-x^2}$. *Expected:* $\int\frac{e^{-x^2}}{x-i\epsilon}\,dx\to \mathrm{PV}\int\frac{e^{-x^2}}{x}\,dx+i\pi f(0)=i\pi$ (the principal value vanishes by oddness). At finite $\epsilon$ the value is $i\pi e^{\epsilon^2}\mathrm{erfc}(\epsilon)$.
7. **(Should)** The Yukawa potential: $\int\frac{d^3k}{(2\pi)^3}\frac{4\pi e^{i\mathbf{k}\cdot\mathbf{x}}}{k^2+\mu^2}$. *Expected:* $\frac{e^{-\mu r}}{r}$. After the angular integral you need $\int_0^\infty\frac{k\sin kr}{k^2+\mu^2}\,dk=\frac{\pi}{2}e^{-\mu r}$ (half of CA08 exercise 7). Connect it to Jackson's screened-potential problem in EM Module 03 and to Proca.
8. **(Should)** The field version. Spatially Fourier transform $\Box G=(\nabla^2-\frac{1}{c^2}\partial_t^2)G=-4\pi\delta^3(\mathbf{x})\delta(t)$. *Expected:* $\ddot{\tilde{G}}+c^2k^2\tilde{G}=4\pi c^2\delta(t)$, so $\tilde{G}_{\mathrm{ret}}(k,t)=4\pi c\,\theta(t)\frac{\sin ckt}{k}$. The angular integral gives $4\pi\frac{\sin kR}{kR}$, so $G=\frac{2c}{\pi R}\theta(t)\int_0^\infty\sin kR\sin ckt\,dk=\frac{c}{R}\delta(R-ct)=\frac{\delta(t-R/c)}{R}$, using $\int_0^\infty\cos(ka)\,dk=\pi\delta(a)$. This is Jackson's retarded Green function.
9. **(Could)** Derive the Kramers–Kronig relations from causality: analyticity in the upper half-plane, Cauchy's formula on a large semicircle with an indentation at $\omega$, and the principal value. *Expected:* $\chi(\omega)=\frac{1}{i\pi}\mathrm{PV}\int\frac{\chi(\omega')}{\omega'-\omega}\,d\omega'$, whose real and imaginary parts are the two relations above.
10. **(Could)** Write one page, in your own words, called "Where the three prescriptions went": the three contours, the three boundary conditions in time, which one allows Wick rotation, and what $G_F$ says about positive and negative frequencies. This is the artifact of the guide.

## Optional extension

Replace $\omega_0$ by $\sqrt{c^2k^2 + m^2c^4/\hbar^2}$ and read off the Klein–Gordon propagator in the time-momentum representation. This is the first line of the Klein–Gordon chapter in a QFT text, and you will have derived it before opening one.

## Pitfalls

- Placing $i\epsilon$ in the wrong place (on $\omega$ versus on $\omega_0^2$ changes the prescription).
- Closing in the wrong half-plane for the sign of $t$.
- Treating Feynman as retarded plus advanced; it is the symmetric part plus a homogeneous solution.
- Forgetting that Wick rotation needs the pole-free quadrants and Jordan's lemma on the arcs.

## Gate questions

1. Why is the integral $\int \frac{e^{-i\omega t}}{\omega_0^2-\omega^2}\, d\omega$ undefined until a prescription is chosen?
2. State the three prescriptions, the pole positions, the half-plane for each sign of $t$, and the boundary condition each encodes.
3. Why is $G_F = \frac{1}{2}(G_{\text{ret}} + G_{\text{adv}})+$ a homogeneous solution?
4. Why does Wick rotation work for the Feynman prescription and fail for the retarded one?
5. Derive $\delta(t - R/c)/R$ from $\tilde{G}(k, t)$.
6. State the chain causality to analyticity to dispersion relation.
7. Where does the principal value come from, and what does the Sokhotski formula add?
8. What changes in the three results when $\omega_0^2 \to c^2k^2 + m^2c^4/\hbar^2$?

Items 1, 2, 4 and 5 are the gate for EM Module 11.

## One-page cheat sheet contents

The table of three prescriptions; the residue values; the half-plane rule; the Wick-rotation criterion; Sokhotski; the Kramers–Kronig pair.

## Where this is used

EM Module 11 (the gate), EM Module 19, EM Module 18 (covariant Green functions), the Zettili scattering and propagator material, and the whole QFT track.

# Appendix: parked material, the QFT outline (Part 2), vocabulary, conventions and tracking

## 1. What this guide leaves out, and the cost of restoring it

All page numbers are Ávila's, from the table of contents you sent.

| Parked | Ávila pages | Why it is parked | Cost to restore | Restore when |
|---|---|---|---|---|
| Multivalued integrands (keyhole contours) | 169 to 173 | Needs branch cuts inside integrals; not used by CA08 or CA09 | about 2 sessions | Part 2, module P1 |
| Logarithmic residues, argument principle | 175 to 178 | Counting zeros and poles; bound states and dispersion theory | about 1 session | Part 2, module P1 |
| Analytic continuation | 181 to 192 | Wick rotation and continuing amplitudes; no effect on EM | about 3 sessions | Part 2, module P3 |
| Riemann surfaces | 193 to 197 | What a branch cut really is | about 1 to 2 sessions | Part 2, module P2 |
| Functions defined by integrals; the gamma function; continuation to the whole plane | 198 to 202 | Dimensional regularization | about 2 sessions | Part 2, module P4 |
| Fluid dynamics | Chapter 7, 204 to 242 | The same mathematics as 2D electrostatics, no new machinery | optional | only for intuition |
| Conformal mapping, invariance of Laplace's equation, electrostatic potential | 245 to 265 (see 248 and 260 to 265) | A second method for the 2D problems of EM Modules 04 to 06 | about 4 sessions | after EM Module 04, as a second route |

Dirichlet and Neumann problems (pp. 114 to 117) are inside CA05 as a "Could" item.

## 2. Part 2 outline: complex analysis for QFT (to be expanded into modules after the CA09 gate)

In Part 1 singularities are obstacles to route around. In Part 2 they carry physics: a pole is a stable particle, a branch point is a production threshold, and their locations are fixed by causality and unitarity rather than by convention. Prerequisite: the CA09 gate. No deadline; the natural slot is the bridge before QFT, or interleaved with the first chapters of Klauber. Each module below becomes a file with the same skeleton as CA01 to CA09.

| # | Module | Source | Sessions | What it becomes in QFT |
|---|---|---|---|---|
| P1 | Multivalued integrands, keyhole contours, the argument principle | Ávila pp. 169 to 178 | 3 | Cuts in propagators; counting bound states |
| P2 | Riemann surfaces | Ávila pp. 187 to 197 | 2 | A cut is a choice, a branch point is physics; thresholds |
| P3 | Analytic continuation and Wick rotation | Ávila pp. 181 to 192 | 3 | Euclidean field theory; continuing amplitudes |

| # | Module | Source | Sessions | What it becomes in QFT |
|---|---|---|---|---|
| P4 | The gamma function and its continuation | Ávila pp. 198 to 202 | 2 | Dimensional regularization: the divergence becomes a pole of $\Gamma$ |
| P5 | Saddle point and stationary phase | **Not in Ávila.** Bender and Orszag, Chapter 6; or Boas | 3 | The semiclassical limit; WKB; instantons. Protect this one: it proves the claim in Mechanics Module 16 that classical mechanics is the stationary-phase approximation to quantum mechanics |
| P6 | Synthesis: analyticity as physics | none | 2 | The singularity dictionary: pole, resonance, branch point, cut; unitarity and the discontinuity across a cut |

One exercise from P3 you can already set up: take your CA09 capstone, the three prescriptions, and rotate the $\omega$ contour by 90°. Which prescription lets you rotate without crossing a pole? (CA09, exercise 4.)

## 3. Vocabulary (Portuguese to English)

Jackson, Churchill and every QFT text use the English terms, so use them from the start.

| Portuguese (Ávila) | English |
|---|---|
| função analítica | analytic (holomorphic) function |
| equações de Cauchy–Riemann | Cauchy–Riemann equations |
| integral de contorno | contour integral |
| teorema de Cauchy | Cauchy's theorem (Cauchy–Goursat) |
| fórmula integral de Cauchy | Cauchy's integral formula |
| série de potências, de Taylor, de Laurent | power series, Taylor series, Laurent series |
| singularidade isolada, removível, essencial | isolated, removable, essential singularity |
| polo (de ordem $m$) | pole (of order $m$) |
| resíduo; teorema do resíduo | residue; residue theorem |
| lema de Jordan | Jordan's lemma |
| valor principal (de Cauchy) | (Cauchy) principal value |
| ramo; ponto de ramificação; corte | branch; branch point; branch cut |
| conexo; simplesmente conexo | connected; simply connected |
| aberto; fechado; fronteira | open; closed; boundary |
| raio de convergência | radius of convergence |
| convergência uniforme | uniform convergence |
| função harmônica; conjugada harmônica | harmonic function; harmonic conjugate |
| princípio do módulo máximo | maximum modulus principle |
| continuação analítica | analytic continuation |
| superfície de Riemann | Riemann surface |
| zero de ordem $m$ | zero of order $m$ |
| contorno indentado | indented contour |
| estimativa $ML$ | ML-estimate |

## 4. Conventions (so that the texts agree)

- **Fourier transform (space):** $\tilde{f}(k) = \int f(x) e^{-ikx}\, dx$ and $f(x) = \int \frac{dk}{2\pi} \tilde{f}(k) e^{ikx}$. This is EM Module 02's convention. Griffiths, Jackson and Figueiredo differ by factors of $2\pi$ and signs; state yours whenever you cross-read.

- **Time dependence:** $e^{-i\omega t}$ (the physics convention, used by Jackson). Then $\tilde{f}(\omega) = \int f(t) e^{i\omega t}\, dt$, and a causal response function is analytic in the **upper** half-plane. Engineering texts use $e^{+j\omega t}$, which moves the analyticity to the lower half-plane. Check which one a source uses before you quote a sign.
- **Orientation:** a contour is positively oriented when it runs counterclockwise (region on the left). Closing in the lower half-plane gives $-2\pi i$ times the residues.
- **Branch of** log**:** principal value with $-\pi < \arg z \leq \pi$ unless the problem says otherwise. State your choice in the first line of any problem that needs one.
- **Greek and symbols:** your Stewart notes use $\partial_u$ for partial derivatives; keep that here.

## 5. Tracking

`notes/error_log.md`**.** One line per stuck step: date, module, exercise, the step ("which half-plane to close in", "residue at a double pole", "sign under clockwise orientation"), what resolved it. If the same step appears three times, redo that module's drill.

**One-line weekly log.** `YYYY-MM-DD, hours, modules/sections, exercises done`.

**Deliverables per module.** A file `notes/caNN-topic.md` with your blank-page derivations in your own hand and a "what surprised me" paragraph, the module's cheat sheet, and worked solutions to exercises 1 to 5.

**Re-entry protocol** (after two or more weeks off): redo the last two derivations of the previous module from a blank page, then continue.

**Verification script.** `tools/verify_expected_results.py` recomputes the numerical and closed-form expected results in the exercises. It needs `numpy`, `scipy`, `mpmath` and `sympy`. If an expected result in a module disagrees with your own calculation, run the script first and tell me which exercise: one of us has a mistake and I want to find it.